

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I M.kasif jalal (192424214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sandeep M

Assessment Tasks - Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A large retail chain operates multiple stores across different cities. Managing inventory manually becomes difficult because products are constantly being sold, restocked, and transferred between branches. To overcome these challenges, the company implements a cloud-based inventory management system that connects all retail stores to centralized cloud services. Every Point-of-Sale (POS) terminal, warehouse system, and management dashboard communicates with cloud servers using APIs. This allows inventory information to be updated instantly whenever a product is sold or received.

The cloud environment enables real-time monitoring, automated stock management, sales analysis, and demand forecasting. By using cloud computing technologies, the retailer improves operational efficiency while reducing infrastructure costs.

Component	Description
Clients	POS terminals, store dashboards, manager web applications
Web APIs	Enable communication between client devices and cloud services
Cloud Services	Inventory management, sales processing, demand forecasting
Cloud Storage	Stores product information, transaction logs, analytics datasets
Platform Services (PaaS)	Analytics engine, machine learning, demand forecasting
Infrastructure (IaaS)	Virtual servers, networking, storage resources

Cloud Service Models

➤ IaaS

- Virtual machines
- Storage
- Networking

➤ PaaS

- Analytics platform
- Forecasting services
- API hosting

➤ SaaS

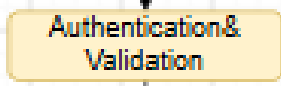
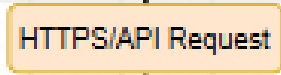
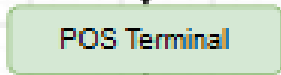
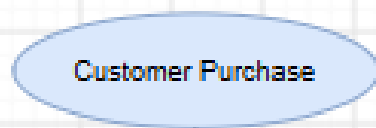
- Inventory dashboard
- Retail management application

Security Features

- VPN for secure store connectivity
- Firewalls
- API Gateway
- Authentication and authorization
- Secure cloud storage
- Network segmentation

Benefits

- Real-time inventory updates
- Centralized management
- Improved demand forecasting
- High scalability
- Reduced infrastructure cost



Case Study 2: Cloud-Based Document Collaboration System

Modern organizations require employees to work together regardless of their physical location. Traditional document sharing methods often create multiple copies of files, making collaboration difficult. A cloud-based document collaboration system solves this problem by allowing users to create, edit, and share documents in real time using web browsers and mobile devices.

Multiple users can simultaneously edit the same document while every change is synchronized instantly across all connected devices. The cloud infrastructure ensures that documents remain available globally with minimal latency.

Component	Description
Clients	Web browsers, laptops, tablets, smartphones
Web APIs	Synchronize document changes
Cloud Storage	Distributed document storage with version control
Platform Services	Real-time collaboration engine
Network Infrastructure	Global content delivery and synchronization

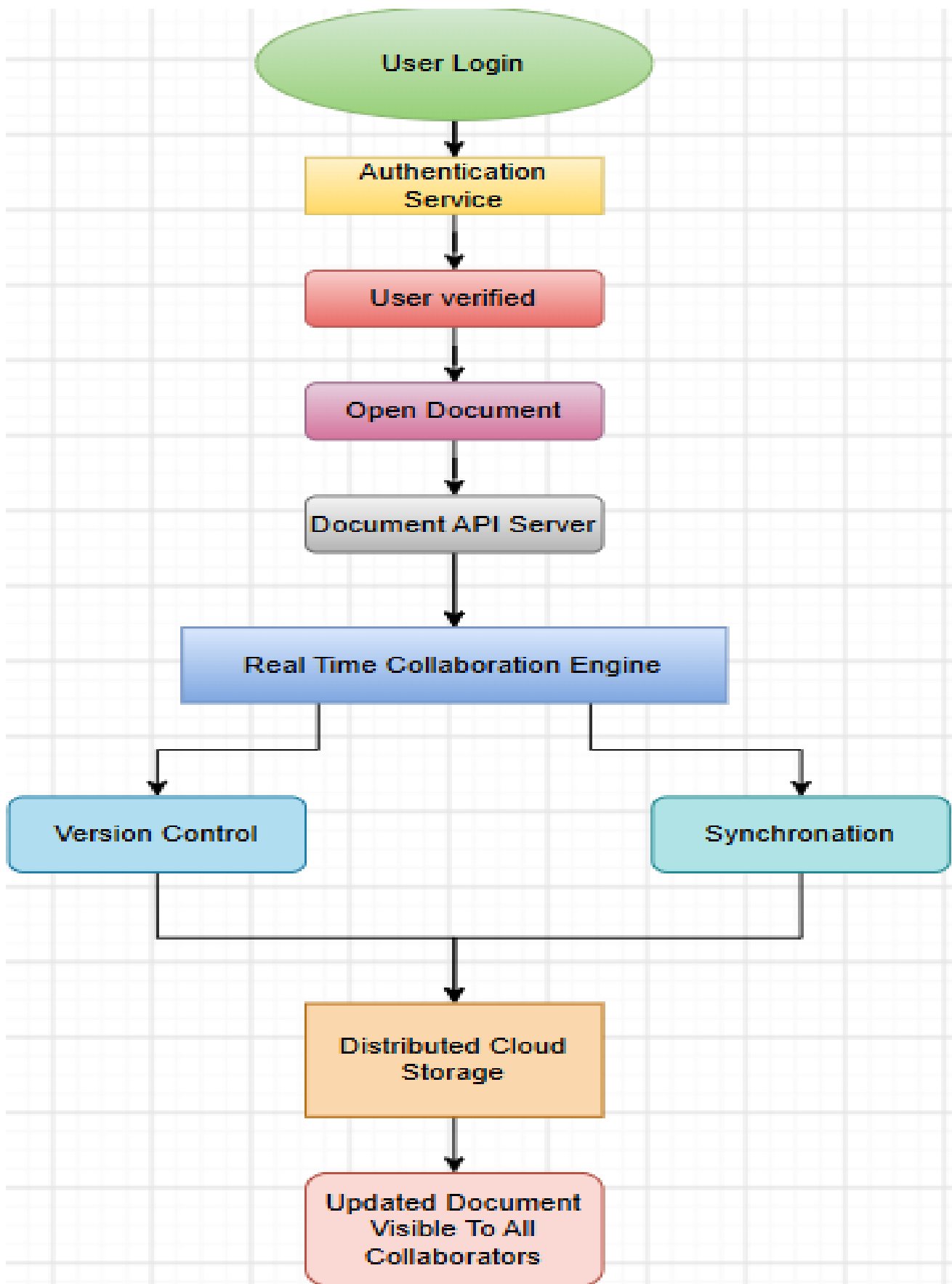
3. Cloud Service Models

- **IaaS**
 - Virtual servers
 - Networking
 - Storage

- **PaaS**
 - Collaboration platform
 - Synchronization services
- **SaaS**
 - Online document editor

Working Process

1. User logs into the application.
2. Authentication server verifies identity.
3. User opens a shared document.
4. Document loads from cloud storage.
5. Multiple users edit simultaneously.
6. Synchronization API updates changes instantly.
7. Version control stores every revision.
8. Changes are replicated across global servers.
9. All collaborators see updates in real time.



Case Study 3: Online Food Ordering Platform

A startup launches a cloud-based food ordering application where customers can browse restaurants, place orders, make payments, and track deliveries. The application uses cloud-native technologies such as RESTful APIs, Kubernetes, virtual machines, cloud databases, and secure authentication for scalable and reliable operations.

System Components

- **Clients**
 - Web Browser
 - Android Application
 - iOS Application
- **Backend Services**
 - Restaurant Service
 - Order Service
 - Payment Service
 - Delivery Service
- **Cloud Platform**
 - Kubernetes Cluster
 - Virtual Machines
 - Load Balancer
- **Cloud Storage**
 - Customer Database
 - Menu Images
 - Invoice Storage

- **Security**
 - HTTPS
 - JWT Authentication
 - Role-Based Access Control (RBAC)

Working Process

1. Customer logs into the application.
2. Customer browses restaurant menus.
3. Menu information is retrieved from cloud storage.
4. Customer places an order.
5. REST API sends the order to backend services.
6. Payment service processes payment.
7. Order is stored in cloud database.
8. Restaurant receives order notification.
9. Delivery partner is assigned.
10. Customer tracks delivery status in real time.

